

A Nonclosed Subspace Counterexample for Enriched $\mathcal{A}$ -Contractions

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Source Problem

The source is Mondal–Garai–Dey, *On some enriched contractions in Banach spaces*, arXiv:2006.11500 [1]. After showing that their fixed-point theorems can fail on arbitrary closed subsets, they ask:

Example 3.4. Choose $X = \mathbb{R}$ with usual norm and take $C = (-\infty, -1] \cup [1, \infty)$. Define $T : C \rightarrow C$ by $Tu = -2u$ for all $u \in C$. Then T is an enriched $\mathcal{A}$ -contraction for $b = \frac{5}{4}$ and $f(r, s, t) = \frac{s+t}{3}$ but T is fixed point free.

In view of Remark 3.3., we pose the following open question:

Open Question 3.5. *Does there exist a self map defined on a subspace (not necessarily closed) of a Banach space satisfying enriched $\mathcal{A}$ -contraction (resp. enriched $\mathcal{A}'$ -contraction) having a fixed point?*

By choosing particular f in above two theorems, we have the following remarks.

Read literally, the question asks only for the existence of at least one fixed-point-bearing example on a subspace. That literal reading has the trivial positive answer given by, for instance, a constant zero map. The contextual question suggested by Remark 3.3 is instead whether the positive fixed-point theorem remains valid when the Banach-space domain is weakened to a linear subspace, not necessarily closed. The following example answers that nontrivial reading negatively, even for an ordinary Banach contraction.

Result

Theorem 1. *There is a real Banach space H , a nonclosed linear subspace $Y \subset H$, and a self-map $T : Y \rightarrow Y$ such that T is both an enriched $\mathcal{A}$ -contraction and an enriched $\mathcal{A}'$ -contraction in the inherited norm, but T has no fixed point in Y .*

Proof Intuition

Completeness is the only missing ingredient. On a Banach space, a strict contraction has its fixed point in the space. On an incomplete normed linear space, the Picard iterates may still converge, but only in the completion. The construction below makes this obstruction visible inside ℓ_2 : each iterate has finite support, while the limit has infinitely many nonzero coordinates.

Proof

Proof. Let $H = \ell_2(\mathbb{N})$, and let $Y = c_{00}$, the subspace of finitely supported real sequences, with the norm inherited from H . Then Y is a linear subspace of the Banach space H , but Y is not closed. Let $(e_n)_{n \geq 1}$ be the standard unit vector basis, and let $R : Y \rightarrow Y$ be the right shift $Re_n = e_{n+1}$. Define

$$Tx = e_1 + \frac{1}{2}Rx, \quad x \in Y.$$

Since R sends finitely supported sequences to finitely supported sequences, T is a self-map of Y .

For $x, y \in Y$,

$$\|Tx - Ty\|_2 = \frac{1}{2}\|R(x - y)\|_2 = \frac{1}{2}\|x - y\|_2.$$

Thus T is an ordinary Banach contraction on the normed space Y .

Now set

$$f(r, s, t) = \frac{1}{2}r, \quad r, s, t \in \mathbb{R}_+.$$

We verify that this single f belongs to both source classes $\mathcal{A}$ and $\mathcal{A}'$. It is continuous and homogeneous, so the corresponding $\mathcal{A}_1, \mathcal{A}_3, \mathcal{A}'_1, \mathcal{A}'_3$ conditions are immediate. For $\mathcal{A}_2$, if

$$r \leq f(s, r, s) = \frac{1}{2}s,$$

then $r \leq ks$ with $k = 1/2$; and if

$$r \leq f(r, s, s) = \frac{1}{2}r,$$

then $r = 0$, again implying $r \leq ks$ with $k = 1/2$. For $\mathcal{A}'_2$, the alternatives are $r \leq r/2$, which forces $r = 0$, or $r \leq s/2$, which gives $r \leq ks$ with $k = 1/2$. The monotonicity condition $\mathcal{A}'_4$ is automatic because f is independent of the third variable. Condition $\mathcal{A}'_5$ says that

$$r \leq f(s, 0, r + s) = \frac{1}{2}s$$

implies $r \leq ks$, again with $k = 1/2$. Finally $\mathcal{A}'_6$ follows from

$$r \leq f(r, r, r) = \frac{1}{2}r,$$

which implies $r = 0$. Hence $f \in \mathcal{A} \cap \mathcal{A}'$.

Taking the enrichment parameter $b = 0$, the enriched $\mathcal{A}$ -inequality becomes

$$\|Tx - Ty\|_2 \leq f(\|x - y\|_2, \|x - Tx\|_2, \|y - Ty\|_2) = \frac{1}{2}\|x - y\|_2,$$

and the enriched $\mathcal{A}'$ -inequality becomes

$$\|Tx - Ty\|_2 \leq f(\|x - y\|_2, \|x - y + y - Ty\|_2, \|y - x + x - Tx\|_2) = \frac{1}{2}\|x - y\|_2.$$

Both hold for all distinct $x, y \in Y$. Thus T is both an enriched $\mathcal{A}$ -contraction and an enriched $\mathcal{A}'$ -contraction.

It remains to show that T has no fixed point in Y . If $x = Tx$, then

$$x = e_1 + \frac{1}{2}Rx.$$

Writing $x = (x_n)_{n \geq 1}$, this gives $x_1 = 1$ and

$$x_n = \frac{1}{2}x_{n-1} \quad (n \geq 2).$$

Therefore $x_n = 2^{-(n-1)}$ for every $n \geq 1$. This sequence belongs to ℓ_2 , but it has infinitely many nonzero coordinates and so is not an element of $c_{00} = Y$. Hence no fixed point of T lies in Y . $\square$

Remark 1. *The Picard iterates from 0 are the finite partial sums*

$$0, \quad e_1, \quad e_1 + \frac{1}{2}e_2, \quad e_1 + \frac{1}{2}e_2 + \frac{1}{4}e_3, \dots,$$

which lie in Y and converge in ℓ_2 to $(1, 1/2, 1/4, \dots) \notin Y$. Thus the example isolates exactly the loss of completeness.

Novelty Check

The bounded search on July 18, 2026 found no duplicate in the run indexes for arXiv:2006.11500, enriched $\mathcal{A}$ -contractions, enriched $\mathcal{A}'$ -contractions, or this subspace fixed-point question. A local source-corpus search found the exact open-question wording only in the source paper and target-pool copies. Web searches for the paper title, the arXiv id, the exact phrase “Does there exist a self map defined on a subspace”, and near variants involving enriched $\mathcal{A}$ - and $\mathcal{A}'$ -contractions found no later explicit resolution.

Human Review Note

The mathematical counterexample is elementary. The main review point is interpretive: it answers the contextual universal fixed-point extension suggested by Remark 3.3, not the literal existential grammar of Open Question 3.5, which is trivially true.

References

- [1] P. Mondal, H. Garai, and L. K. Dey, *On some enriched contractions in Banach spaces*, arXiv:2006.11500, 2020.